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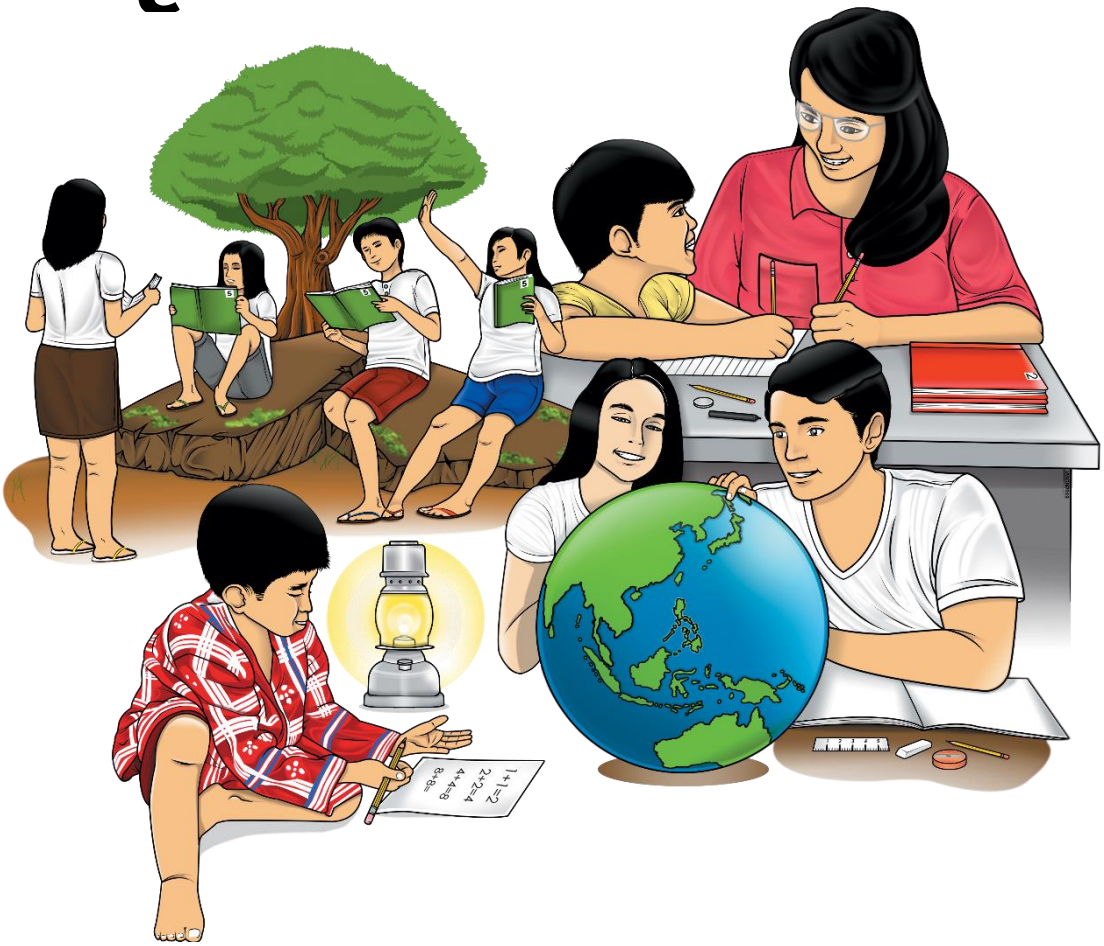
General Chemistry 1

Quarter 2 - Module



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Quarter 2 - Module
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What I Need to Know

This module was designed and written with you in mind. It is here to help you master the topic about quantum numbers. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course, but the order in which you read them can be changed to correspond with the textbook you are now using.

At the end of this lesson, you are expected to:

Use quantum numbers to describe an electron in an atom. (STEM_GC11ESIIa-b-54)

What I Know

Directions: Encircle the letter of the correct answer.

1. Which quantum number represent a specific orbital in an atom?
   A. *n* B. *l* C. *ml* D. *ms*
2. How many electrons are there in first energy level (*n*=1)?
   A. 1 B. 2 C. 8 D. 0
3. What is the value of the azimuthal quantum number, *l*, for the d-subshell?
   A. 0 B. 1 C. 2 D. 3
4. Which of the following refers to spin quantum number (*ms*)?
   A. the energy of electron
   B. direction of electron spin
   C. orientation of orbitals around the nucleus
   D. sublevel or shape of the orbital in the electron
5. What will be the principal quantum number of an electron in an f sublevel?
   A. 1 B. 2 C. 3 D. 4

Lesson 1 Quantum Numbers

What's New

Directions: Label each of the following sets of quantum numbers as either *Valid*, which indicates that sets is a legitimate set of quantum numbers for an electron, or *Invalid*, which would indicate that the set is not a possible set of quantum numbers for an electron.

n	ℓ	m_ℓ	m_s	Valid or Invalid?
1	0	0	½	
1	1	+1	-½	
2	0	0	-½	
2	2	-2	½	
2	1	-1	½	
3	2	-1	½	
3	1	0	0	
3	0	1	½	
3	0	0	-½	
1	2	-3	½	

What Is It

Each electron in an atom is described by four different quantum numbers. The first three (*n, l, ml*) describe the size, shape, and orientation in space of the orbitals on an atom. and the fourth (*ms*) specifies how many electrons can occupy that orbital. These four quantum numbers are used to describe the probable location of an electron in an atom.

Quantum Numbers

1. Principal Quantum Number (*n*)

- It describes the **main energy level (MEL)** on which the orbital resides.
- It describes the general distance of an electron from the nucleus (the farther the electron from the nucleus, the higher is the energy).
- It largely determines the energy of an atom.
- It can be any nonzero positive integer: 1, 2, 3, 4……

Example No. 1:

If n=7, what is the principal electron shell?

Answer: Since n refers to the principal energy level/ or energy shell, so if n=7 then the principal energy shell =7.

2. Azimuthal Quantum Number (*l*)

- Also called **angular momentum number**
- It describes the subshell/sub-energy level (SEL) where the electron is
- Allowed values of l are integers ranging from 0 to $n - 1$. Thus, for a given value of n, there are different possible values of l.

If *n* equals	***l* can be**
1	0
2	0 or 1
3	0,1, or 2
4	0, 1, 2, or 3

- Commonly, instead of referring to the numerical value of l, a letter represents the value of l (to help distinguish it from the principal quantum number).

If *l* equals	**The letter is**
0	s
1	p
2	d
3	f

- It affects the spatial distribution of the electron in three-dimensional space; that is, the shape of the electron's distribution in space.

l	**Sublevel**	**Orbital Shape**
0	s	Spherical
1	p	Dumbbell shaped
2	d	4 cloverleaf-shaped
3	f	(Too complex to describe)

Note: s = sharp; p = principal; d = diffuse; f = fundamental

Example No. 2:

If n=7, what are the possible values of l?

Answer: Since l can be zero or a positive integer less than (n-1), it can have a value of 0, 1, 2,3,4,5, or 6.

3. Magnetic Quantum Number (*ml*)

- Describes the three-dimensional orientation of an electron's distribution in space
- Allowed values of ml are integers ranging from −l to l: $-l \leq ml \leq l$

- Therefore, in any given energy level, there can be up to 1 s orbital, 3 p orbitals, 5 d orbitals, 7 f orbitals, and so forth.

If *l* equals	The *ml* values can be
0	0
1	-1, 0 or 1
2	-2, -1, 0, 1 or 2
3	-3, -2, -1, 0, 1, 2 or 3

The value of *ml* dictates the orientation of an electron's distribution in space. When *l* = zero, *ml* can be only zero, so there is only one possible orientation. When *l* = 1, there are three possible orientations for an electron's distribution. When *l* = 2, there are five possible orientations of electron distribution. This goes on and on for other values of *l*, but we need not consider any higher values of *l* here. Each value of *ml* designates a certain orbital. Thus, there is only one orbital when *l* is zero, three orbitals when *l* is 1, five orbitals when *l* is 2, and so forth.

Example No. 3:

If *n*=3 and *l*=2, then what are the possible values of *ml*?

Answer: Since *ml* must range from –*l* to +l, then *ml* can be -2, -1, 0, 1, or 2.

4. Spin Quantum Number (*ms*)

- Specifies the orientation of the spin axis of an electron.
- An electron can spin in only one of two directions (sometimes called up and down)
- Can be +1/2 or -1/2

Example No. 4:

Can an electron with *ms* = ½ have a downward spin?

Answer: No, if the value of *ms* is positive, the electron is "spin up".

To summarize the four quantum numbers, refer to the table below:

Name	Symbol	Permitted Values	Property
principal	n	positive integers(1,2,3, ...)	orbital energy (size)
angular momentum	*l*	integers from 0 to n-1	orbital shape (The *l* values 0, 1, 2, and 3 correspond to s, p, d, and f orbitals, respectively.)
magnetic	m_l	integers from -*l* to 0 to +*l*	orbital orientation
spin	m_s	+1/2 or -1/2	direction of e^- spin

What's More

Examples:

Element	Atomic Number	Electron Configuration	*n*	*l*	*ml*	*ms*
H	1	$1s^1$	1	0	0	1/2
He	2	$1s^2$	1	0	0	-1/2
Li	3	$1s^22s^1$	2	0	0	1/2
Be	4	$1s^22s^2$	2	0	0	-1/2
B	5	$1s^22s^22p^1$	2	1	-1	1/2
C	6	$1s^22s^22p^2$	2	1	0	1/2
N	7	$1s^22s^22p^3$	2	1	+1	1/2
O	8	$1s^22s^22p^4$	2	1	-1	-1/2

Directions: Given the following examples, provide the quantum numbers of the given elements.

1. Magnesium
2. Sulfur
3. Manganese
4. Selenium
5. Cobalt

What I Can Do

Directions: A. For each set of quantum numbers, state which one is invalid and why.

1. n = 1, l = 1 ml = 0 ms = +1/2

_____

2. n = 2, l = 1 ml = -1 ms = +3/2

_____

3. n = 3, l = 1 ml = 2 ms = -1/2

_____

B. State the number of possible electrons described by the following quantum numbers.

4. n = 3, l = 1 _____
5. n = 3, l = 2, ml = -1 _____

Assessment

Directions: Fill in the blanks with the correct response.

1. The number of electrons with the quantum numbers n=3, l=2 and ml= 0 is _____.
2. The subshell with the quantum numbers n=4, l=2 is _____.
3. The ml values for a d orbital are _____.
4. The allowed values of l for the shell with n=2 are _____.
5. The allowed values of l for the shell with n=4 are _____.

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the topic magnetic property of atoms. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course, but the order in which you read them can be changed to correspond with the textbook you are now using.

At the end of this lesson, you are expected to:

Determine the magnetic property of the atom based on its electronic structure. (STEM_GC11CB-IIa-b-57)

What I Know

Directions: Encircle the letter of the correct answer.

1. What is the total number of electrons that can occupy the principal energy level n?
   A. 2 B. 8 C. n D. 2n2
2. Calcium has an atomic number of 20. A stable calcium atom has an electronic configuration of:
   A. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$
   B. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^2$
   C. $1s^2 2s^2 2p^6 3s^2 3p^6$
   D. $1s^2 1p^6 2s^2 2p^6 3s^2 3p^2$
3. The electrons with principal energy level n = 2 of a stable atom of boron (atomic number of 5) have an electron arrangement of:
   A. (↑ ↓) (↑) ( ) ( )
   B. (↑) (↑) (↑) ( )
   C. ( ) (↑) (↑) (↑)
   D. (↑ ↓) (↑ ↓) (↑) ( ) ( )
4. Which of the below explains why the 4s sub-level fills before the 3d sub-level?
   A. Aufbau Principle
   B. Hund's Rule
   C. Octet Rule
   D. Pauli's Exclusion Principle

5. The electron configuration of an atom is $1s^2 2s^2 2p^6$. The number of valence electrons in the atom is

A. 3 b. 6 C. 8 D. 10

Lesson 2 Magnetic Property

What's New

Directions: Rearrange the letters of the correct term that is described by the corresponding statement.

_____ 1. LAPNICRIP – It is the quantum number that represents the main energy level of the electron.

_____ 2. LATOBRI – It is a representation of the wave function of a hydrogen-like atom.

_____ 3. ALGANUR MUTMENMO – It is a quantum number that represents the shape of orbitals.

_____ 4. NOTRECLE – It is the particle that can be described by four quantum numbers

_____ 5. MEGATINC – It represents the quantum number that describes the orientation of an orbital.

What Is It

One of the properties of an element that can be determined using electron configuration is the **magnetic property**. **Magnetism** is the behavior of a substance when exposed to a magnetic field that is related to the presence of paired and unpaired electrons in the material. It can be predicted using an orbital diagram. **Paramagnetic substances** are those that have one or more unpaired electrons and can be attracted to a magnet. On the other hand, the **diamagnetic substance** is slightly repelled by a magnet due to the presence of only paired electrons in its orbitals.

A useful general rule that an atom with odd number of valence electrons must be paramagnetic since we need an even number of valence electrons for complete pairing. On the other hand, an atom with even number of electrons may be either paramagnetic or diamagnetic.

What's More

Directions: Fill in the table with necessary information.

Element	Atomic Number	Electron Configuration	Number of Unpaired Electrons	Magnetic Property
Al	13			
Br	35			
Au	79			
Mo	42			
Pb	82			

What I Can Do

Directions: Indicate the number of unpaired electrons and identify the magnetic property present in each of the following atoms:

a. B _____ f. Se _____

b. Ne _____ g. Kr _____

c. P _____ h. Cd _____

d. Sc _____ i. I _____

e. Mn _____ j. Pb _____

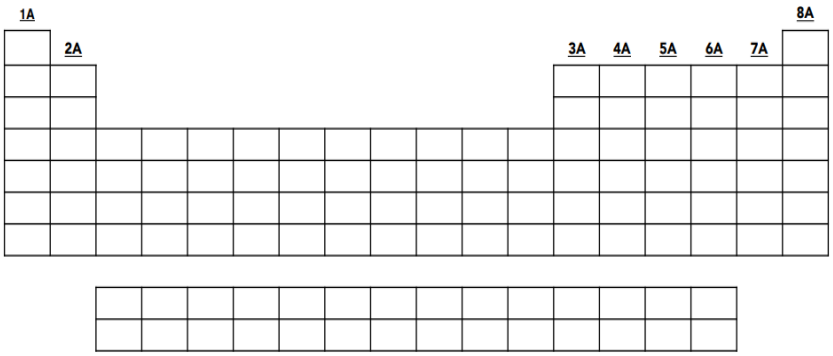
Assessment

Directions: Answer each of the questions below.

1. Given the following quantum numbers, determine the element, write the electron configurations, and identify the magnetic property.

Element	Quantum Numbers	Electron Configuration	Magnetic Property
	n=1, *l*= 0, *ml*= 0, *ms*=-1/2		
	n=2, *l*=1, *ml*= 0, *ms*= +1/2		
	n=3, *l*=1, *ml*= 0, *ms*= +1/2		
	n=4, *l*=2, *ml*= 0, *ms*= +1/2		
	n= 6, *l*=0, *ml*= 0, *ms*= -1/2		

2. Plot the elements in number 1 in the blank periodic table provided for.



What I Need to Know

This module was designed and written with you in mind. It is here to help you master the topic about orbital diagram. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course, but the order in which you read them can be changed to correspond with the textbook you are now using.

At the end of this lesson, you are expected to:

Draw an orbital diagram to represent the electronic configuration of atoms. (STEM_GC11CB-IIa-b-58)

What I Know

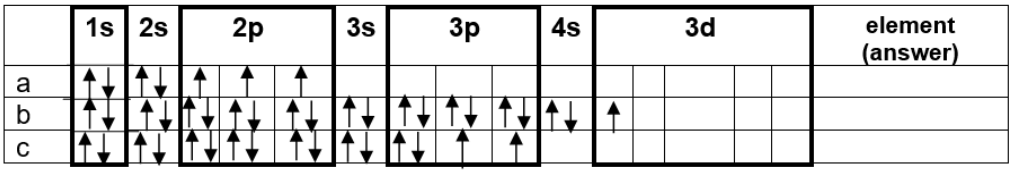
Directions: Match column A to column B.

A	B
_____1. It states that electrons will occupy the orbitals having lower energies before occupying higher energy orbitals.	A. Hund's Rule
_____2. It states that no two electrons in the same atom can have the same set of four quantum numbers.	B. Aufbau Principle

Lesson 3 Electron Configuration and Orbital Diagram

What's New

Directions: Identify the element that has the following orbital diagram.



What Is It

Electron Configuration

The electron configuration is defined as the probable distribution of electrons in their atomic orbitals. It is a simple way of writing down the locations of all the electrons in an atom. We shall be mainly concerned with the electron configuration in the ground state (lowest energy state of an atom) since that is the most stable arrangement.

To determine the lowest energy arrangement of the electrons, we shall follow three rules:

1. Aufbau Principle

- This principle is named after the German word 'Aufbau' which means 'build up'.
- The Aufbau principle states that electrons will occupy the orbitals having lower energies before occupying higher energy orbitals.
- According to this principle, electrons are filled in the following order: 1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p, 7s, 5f, 6d, 7p…
- The order in which electrons are filled in atomic orbitals as per the Aufbau principle is illustrated at the right side.

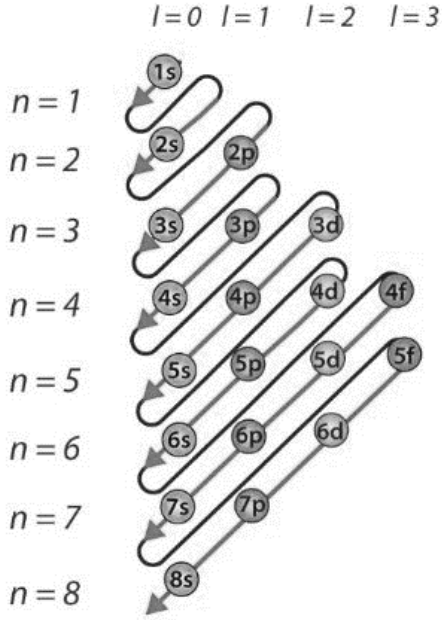
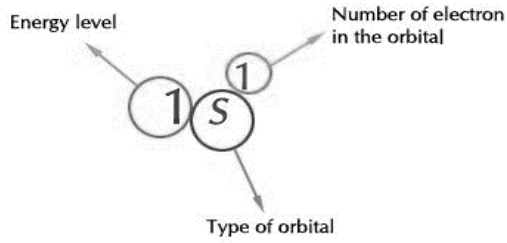


Figure 1: Technique for determining the correct order of filling the orbitals.

The electron configuration shows how the electrons of an atom are distributed among the atomic orbitals. Say for example the electron configuration of Hydrogen is:



Here are other examples of how to write the electron configuration of some elements.

$_{3}Li^{7}$ =$1s^2 2s^1$

$_{6}C^{12}$ = $1s^2 2s^2 2p^2$

$_{11}Na^{23}$ = $1s^2 2s^2 2p^6 3s^1$

Noble Gas Configuration

The standard notation often yields a long electron configuration (especially for elements having a relatively large atomic number). In such cases, an abbreviated or condensed notation may be used instead of the standard notation. An abbreviated notation consists of the symbol of the last noble gas before that atom, followed by the configuration of the remaining electrons.

Examples:

Element	Electron Configuration	Noble Gas Configuration
$_{7}N^{14}$	$1s^2 2s^2 2p^3$	[He] $2s^2 2p^3$
$_{13}Al^{27}$	$1s^2 2s^2 2p^6 3s^2 3p^1$	[Ne] $3s^2 3p^1$
$_{18}Ar^{40}$	$1s^2 2s^2 2p^6 3s^2 3p^6$	[Ne] $3s^2 3p^6$

Important data that can be gathered from the shortened electronic configuration are the following:

Element	Electron Configuration	Noble Gas Configuration
$_{7}N^{14}$	$1s^2 2s^2 2p^3$	[He] $2s^2 2p^3$
$_{13}Al^{27}$	$1s^2 2s^2 2p^6 3s^2 3p^1$	[Ne] $3s^2 3p^1$
$_{18}Ar^{40}$	$1s^2 2s^2 2p^6 3s^2 3p^6$	[Ne] $3s^2 3p^6$

a. Valence configuration: The electronic configuration representing the outermost subshells.
b. Valence electrons: the number of electrons in the outermost subshells.

Exceptions

It is important to note that there exist many exceptions to the Aufbau principle such as chromium and copper. These exceptions can sometimes be explained by the stability provided by half-filled or completely filled subshells.

The electron configuration of Chromium (Z=24) is [Ar] $4s^1 3d^5$ and not [Ar] $4s^2 3d^4$, as we might expect. A similar case is observed for Copper, whose electron configuration is [Ar] $4s^1 3d^{10}$ rather than [Ar] $4s^2 3d^9$. The reason for these irregularities is that a slightly greater stability is associated with the half-filled ($3d^5$) and completely filled ($3d^{10}$) subshells.

2. Pauli Exclusion Principle

- The Pauli Exclusion Principle states that no two electrons in the same atom can have the same set of four quantum numbers.
- Consequently, it means that a maximum of two electrons, each having opposite spins, can fit in an orbital.
- Therefore, if the principal, azimuthal, and magnetic numbers are the same for two electrons, they must have opposite spins.

3. Hund's Rule

- This rule describes the order in which electrons are filled in all the orbitals belonging to a subshell.
- It states that every orbital in a given subshell are singly occupied by electrons before a second electron is filled in an orbital.
- To maximize the total spin, the electrons in the orbitals that only contain one electron all have the same spin (or the same values of the spin quantum number).

  Consider the correct electron configuration of the Nitrogen (Z = 7) atom: $1s^2\ 2s^2\ 2p^3$

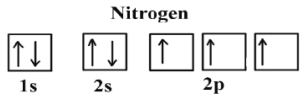


Figure 2: Orbital diagram of Nitrogen.

The p orbitals are half-filled; there are three electrons and three p orbitals. This is because the three electrons in the 2p subshell will fill all the empty orbitals first before pairing with electrons in them. Next, consider oxygen (Z = 8) atom, the element after nitrogen in the same period; its electron configuration is: $1s^2\ 2s^2\ 2p^4$

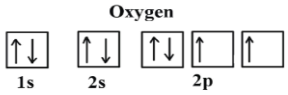


Figure 3: Orbital diagram of Oxygen.

Oxygen has one more electron than nitrogen; as the orbitals are all half-filled, the new electron must pair up.

What's More

Directions: A. Complete the table below. Supply the missing information.

Element	Atomic Number	No. of e^-	Electron Configuration	Orbital Diagram	No. of Valence e^-
Na			Conventional		
			Noble Gas Configuration		

V			Conventional Noble Gas Configuration		
Cl			Conventional Noble Gas Configuration		
Zn			Conventional Noble Gas Configuration		
Si			Conventional Noble Gas Configuration		

What I Can Do

Directions: A. The ground state electron configurations listed here are incorrect. Explain what mistakes have been made in each and write the correct electron configurations.

a. Al : $1s^2 2s^2 2p^4 3s^2 3p^3$

_____

b. B: $1s^2 2s^2 2p^5$

_____

c. F: $1s^2 2s^2 2p^6$

_____

d. K: $1s^2 2s^2 2p^6 3s^2 3p^6 4d^1$

_____

e. Cl: $1s^2 2s^2 2p^6 3s^2 3p^3$

_____

Assessment

Directions: Encircle the letter of the correct answer.

1. Which neutral atom has the following configuration: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^6\ 5s^2\ 4d^9$?
   A. gold B. silver C. copper D. cadmium
2. What is the proper Noble Gas notation for the electron configuration of a neutral atom of Germanium (Ge)?
   A. [Ar] $4s^2 4d^{10} 4p_2$ B. [Kr] $4s^2 3d^{10} 4p^2$ C. [Ne] $4s^2 3d^{10} 4p^2$ D. [Ar] $4s^2 3d^{10} 4p^2$
3. Which neutral atom has the following electron configuration: [Xe] $6s^2\ 5d^4\ 4f^{14}$?
   A. hassium (Hs) B. lutetium (Lu) C. rhenium (Re) D. tungsten (W)
4. Which of the following explains why the 4s sub-level fills before the 3d sub-level?
   A. Octet Rule C. Aufbau Principle
   B. Hund's Rule D. Pauli Exclusion Principle
5. Most exceptions to the electron-filling order indicated by an arrow diagram occur as a result of _____.
   A. the special stability of a half-filled sublevel.
   B. the stability of eight electrons in the outer level.
   C. the increased electron capacity of higher energy levels.
   D. the increased energy associated with higher principal quantum numbers.

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the topic about chemical bonding. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course, but the order in which you read them can be changed to correspond with the textbook you are now using.

At the end of this lesson, you are expected to:

1. **Draw the Lewis structure of ions. (STEM_GC11CBIId-g-70)**
2. **Apply the octet rule in the formation of molecular covalent compounds. (STEM_GC11CBIId-g-76)**
3. **Write the formula of molecular compounds formed by the nonmetallic elements of the representative block. (STEM_GC11d-g-77)**

What I Know

Directions: Fill in the K-W-H-L Chart below about chemical bonding.

K	W	H	L
What do I know?	What do I want to find out?	How can I found out what I learn?	What did I learn?

Lesson 4 Chemical Bonding

What's New

Directions: Illustrate how NaCl (sodium chloride) was formed through ionic bonding and HCl (hydrochloric acid) through covalent bonding.

NaCl (sodium chloride)	HCl (hydrochloric acid)

What Is It

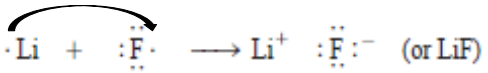
When atoms interact to form a chemical bond, only their outer regions are in contact. For this reason, when we study chemical bonding, we are concerned primarily with the valence electrons of the atoms. To keep track of valence electrons in a chemical reaction, and to make sure that the total number of electrons does not change, chemists use a system of dots devised by Lewis and called Lewis's dot symbols. A **Lewis dot symbol** consists of the symbol of an element and one dot for each valence electron in an atom of the element. All elements belonging to the same group in the periodic table have the same number of valence electrons, hence the same number of dots around their chemical symbols for their Lewis dot symbols.

Ionization energy is the minimum amount of energy required to remove an electron from a gaseous atom in its ground state. It increases from left to right across a period. Elements with small ionization energy tend to easily give up electrons to become positive ions or cations. **Electron affinity** of an atom is the energy change that occurs when a gaseous

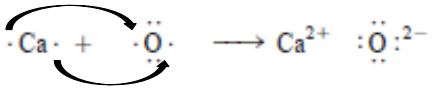
atom accepts an electron. It increases from left to right across a period. The greater the electron affinity, the greater the tendency for an atom to accept an electron and form a negative ion or anion. Atoms with low ionization energies tend to form cations, while those with high electron affinities tend to form anions. These cations and anions combine to form ionic compounds. As a metal transfers electrons to a nonmetal, an ionic bond is formed. An **ionic bond** is an electrostatic force that holds ions together in an ionic compound.

Using the Lewis dot symbols of the representative elements, how ionic compounds form from combinations of these elements can be shown as follows.

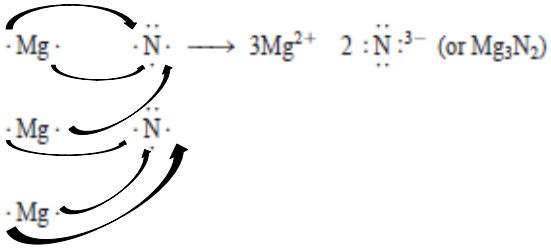
a. combining lithium from Group 1A and fluorine from Group 2A,



b. combining calcium from Group 2A and oxygen from Group 6A,

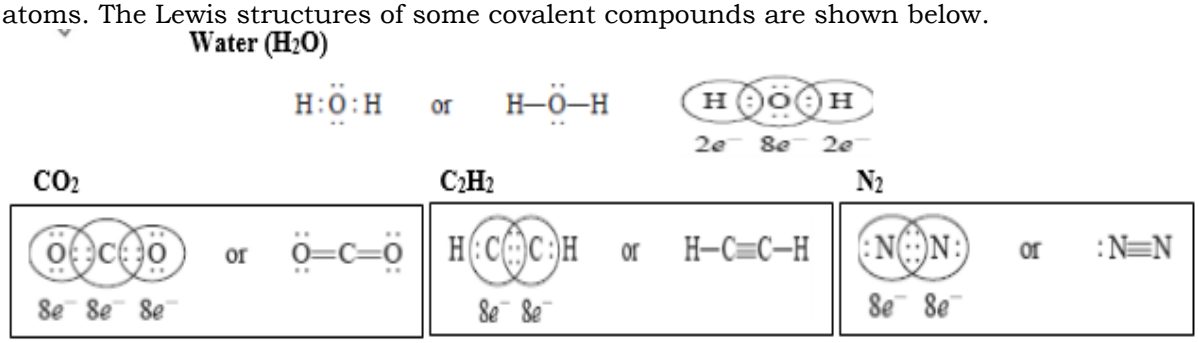


c. combining magnesium from Group 2A and fluorine from Group 5A,



The **Octet Rule** states that when atoms of elements combine to form compounds, they tend to lose, gain, or share electrons to achieve the same electron arrangement as the noble gas nearest them in the periodic table. When two electrons are shared equally between two atoms, a covalent bond is formed. A **covalent bond** is a type of bond in which two electrons are shared by two atoms. For simplicity, the shared pair of electrons is often represented by a single line -.

Atoms can form different types of covalent bonds: a **single bond** is formed when two atoms share a pair of electrons, **double bond** when two pairs of electrons are shared, and **triple bond** when three pairs of electrons are shared. Carbon, nitrogen, oxygen, and sulfur are capable of multiple bonds. The structures we use to represent these covalent bonds, showing only valence electrons, are called Lewis structures. A **Lewis structure** is a representation of covalent bonding in which shared electron pairs are shown either as lines or as pairs of dots between two atoms, and lone pairs are shown as pairs of dots on individual atoms. The Lewis structures of some covalent compounds are shown below.



In diatomic molecules in which atoms are identical such as H_2, we expect equal sharing of electrons, i.e., electrons spend the same time in the vicinity of each atom. This is a **non-polar covalent bond**. In molecules with different atoms such as HF, atoms do not share electrons equally. This bond is called **polar covalent bond** because electrons spend more time in the vicinity of one atom than the other.

Electronegativity is the ability of an atom to attract toward itself the electrons in a chemical bond. This property of an atom tells if an atom can form a covalent molecule or compound with another atom.

In writing chemical formulas of molecular compounds formed by the nonmetallic representative elements in the periodic table, the following are a few guidelines.

Guideline #1: State the name of the first element. The second element ends in (suffix) –ide.

Examples: HF - hydrogen fluoride; SiC - silicon carbide

Guideline #2: Prefixes are used to denote the number of atoms of each element in the formula. The prefix 'mono' is usually omitted for the first element. The second element ends in (suffix) –ide.
Examples: CO -carbon monoxide; N_2O_4 – dinitrogen tetroxide

What's More

Directions: A. Complete the following table.

Element	Atomic number	Electron configuration	Group number (Old)	Period number	Number of valence electrons	Lewis dot symbol
Li						
Mg						
Al						
C						
O						

B. Use Lewis dot symbols to show the formation of the following ionic compounds.

Ionic Compound	Lewis Dot Symbol (Formation)
1. Aluminum oxide	
2. Barium hydride	
3. Magnesium sulfide	
4. Potassium nitride	
5. Calcium chloride	

C. Identify the chemical name of the following compounds given the chemical formulas.

Chemical Formula	Chemical Name
OF_2	1.
Si_3N_4	2.
ClF_3	3.
$SbCl_5$	4.
N_2H_4	5.

What I Can Do

Directions: Do the following.

Look for ingredients in some commercial products. In tabulated form, give the names of the products, the chemical names, the chemical formulas of the ingredients, and the type of chemical bond/s exist/s between the atoms.

Assessment

Directions: Encircle the letter of the best answer.

1. Which atom has five valence electrons in its Lewis dot symbol?
   A. S B. Cl C. A1 D. Ca
2. How many electrons must a nitrogen atom gain to attain a noble-gas electron configuration?
   A. 1 B. 3 C. 5 D. 7
3. Which will form a cation when it loses valence electron/s?
   A. sulfur B. sodium C. fluorine D. phosphorus
4. Which atom must gain two electrons to achieve a stable electron configuration?
   A. N B. S C. P D. Cl
5. Which is NOT isoelectronic with neon?
   A. Cl^- B. O^{2-} C. N^{3-} D. C^{4-}

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the topic about Lewis structure of molecular covalent compounds. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course, but the order in which you read them can be changed to correspond with the textbook you are now using.

At the end of this lesson, you are expected to: **Draw the Lewis structure of molecular covalent compounds. (STEM_GC11BIId-g-78)**

What I Know

Trivia: Did you know that?

The world of chemical science was introduced to the **principle of covalence** in 1919. Future Nobel Prize-winning chemist **Irving Langmuir** coined the term to describe the molecular bonds formed by electrons in the outermost shell or valence of atoms. The term "**covalent bond**" first came into use in 1939.

Lesson 5 Lewis Structure of Molecular Covalent Compounds

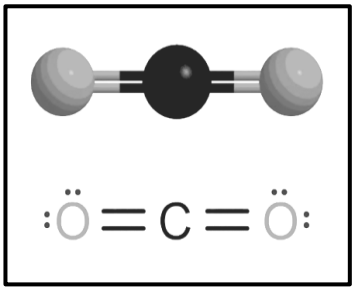
What's New

Directions: Fill in the blanks with the correct words to complete the paragraph below.

Covalent bonds are characterized with _____ of electrons between two or more atoms. These bonds mostly occur between _____ or between two same _____. Due to the difference in _____ of atoms, one cannot _____ electrons so the atoms instead share electrons to satisfy the _____ configuration which is also known as the "_____". The _____ properties of covalent compounds such as _____ point, boiling point, and electrical _____ are affected because of the sharing in the electrons of atoms.

What Is It

Covalent bonding involves sharing of valence electrons among atoms to achieve octet configuration. The **physical properties** of covalent compounds such as melting point, boiling point, and electrical conductivity are affected by the sharing in the electrons of atoms. Methane consists of one atom of carbon and four atoms of hydrogen. Four covalent bonds are formed so that C has an octet of valence electrons, and each H has two valence electrons - one from the carbon atom and one from one of the hydrogen atoms. Covalent compounds that are more complex and consist of multiple atoms can also be represented through Lewis-dot structure.



https://www.thoughtco.com/definition-of-lewis-structure-605306

What's More

Directions: Follow the procedure in the activity below.

Activity: Drawing the Lewis Structure of Common Covalent Compounds
Materials: Worksheet, Periodic Table of Elements, Paper, Pen

Procedure: A. In the set of compounds inside the box below, identify and encircle the covalent compounds.

CaF_2	Na_2O	BaF_2	PCl_5	Al_2O_3	CS_2
Cl_2	$BaCl_2$	H_2O	CH_4	HF	$MgCl_2$
HCl	Mg_3N_2	$AlCl_3$	CO_2	I_2	Na_3N

B. Draw the Lewis dot structure of the covalent compounds that you identified in Part A.

What I Can Do

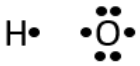
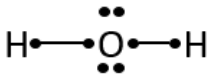
Directions: Read and explain the question below.

How does the sharing of electrons in covalent bonds affect the physical and chemical properties of covalent compounds?

_____

Assessment

Directions: Write the chemical formula of the following covalent compounds and draw their Lewis-dot structure.

Covalent Compound	Chemical Formula	Lewis Dot Symbol	Lewis Dot Structure
Dihydrogen monoxide	H_2O		
1. Boron trifluoride			
2. Phosphorus tribromide			
3. Sulfur dioxide			
4. Dinitrogen monoxide			
5. Tetrabarium hexachloride			

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the topic about geometry of simple compounds. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course, but the order in which you read them can be changed to correspond with the textbook you are now using.

At the end of this lesson, you are expected to:
Describe the geometry of simple compounds. (STEM_GC11BIId-g-81)

What I Know

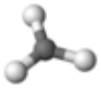
Directions: Fill in the K-W-H-L Chart below about molecular geometry of simple compounds.

K	W	H	L
What do I know?	**What do I want to find out?**	**How can I found out what I learn?**	**What did I learn?**

Lesson 6 Molecular Geometry of Simple Compounds

What's New

Directions: Match column A to column B.

Molecular Shape	Molecular Figure
_____1. Linear	a.
_____2. Bent	b.
_____3. Trigonal planar	c.
_____4. Pyramidal	d. 
_____5. Tetrahedral	e.

What Is It

In determining the molecular geometry of a compound, one must know the Lewis-dot structure. It is the first step to predict the shape of molecules. It is also important to know the concept of **Valence-Shell Electron-Pair Repulsion (VSEPR)** Theory which states that electron pairs repel each other. Thus, electron pairs will spread themselves as far from each other as possible to minimize repulsion.

Bonding electron pairs	Lone pairs	Electron domains (Steric #)	Shape	Ideal bond angle (example's bond angle)	Example	Image
2	0	2	linear	180°	CO_2	
3	0	3	trigonal planar	120°	BF_3	
2	1	3	bent	120° (119°)	SO_2	
4	0	4	tetrahedral	109.5°	CH_4	
3	1	4	trigonal pyramidal	107°	NH_3	

The VSEPR model focuses on the bonding and non-bonding electron pairs present in the valence shell of an atom that connects with two or more other atoms. The molecular structure is determined through the use of a steric number. It is the number of bonding electron pairs together with the number of lone pairs attached to the central atom. The bond

angle is also taken into consideration in molecular geometry since there is repulsion in the electrons of atoms.

In the table are the common molecular geometry of simple compounds. In a **linear model**, atoms are connected in a straight line, and a bond angle is simply the geometric angle between two adjacent bonds. A simple triatomic molecule of type AX_2 has its two bonding orbitals 180° apart. Molecules with the **trigonal planar shape** are triangular and in one plane.

An AX_3 molecule has three regions of electron density extending out from the central atom. The repulsion between these will be at a minimum when the angle between any two is 120°. Tetra- signifies four, and -*hedral* relates to a face of a solid; "tetrahedral" literally means "having four faces." This shape is found when there are four bonds all on one central atom, with no lone electron pairs. By the VSEPR theory, the bond angles between the electron bonds are 109.5°.

What's More

Directions: Follow the procedure below.

Activity: Ball & Stick Model of a Simple Compound
Materials: Spherical objects (i.e., ping pong balls, clay, styrofoam), stick (i.e., barbecue stick, straw), glue stick (or any adhesive objects like glue, paste, adhesive tape)
Procedure: A. Complete the table.

Compound	Steric Number	Molecular Geometry
1. $BeCl_2$		
2. BF_3		
3. NH_4		
4. PCl_5		
5. SF_6		

Part B: Give the steric number and molecular geometry of the following compounds.

1. Choose one compound from the five compounds inside the table. Using your materials create a ball and stick model of the molecular geometry of your chosen compound.
2. Spherical objects represent the atoms of elements in a compound, the same atoms should have the same color. The sticks on the other hand represent the bonds between the atoms.
3. Refer to the table given in what is it section as your guide in creating an accurate representation of molecular geometry.
4. Take a picture of your work together with important details such as the name of the compound and its molecular geometry and background information about your chosen compound. Submit the picture of your work to your respective subject teacher.

What I Can Do

Directions: Read and answer the question below.

How does the molecular geometry of compounds affect its properties? State a specific situation or example.

_____

Assessment

Directions: Provide two example of compounds that exhibit the following molecular structure.

Molecular Structure	Compounds
Bent	1. 2.
Linear	3. 4.

Tetrahedral	5. 6.
Trigonal pyramidal	7. 8.
Trigonal planar	9. 10.

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the topic about polarity of simple molecules. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course, but the order in which you read them can be changed to correspond with the textbook you are now using.

At the end of this lesson, you are expected to:
Determine the polarity of simple molecules. (STEM_GC11BIId-g-82)

What I Know

Directions: Write TRUE if the statement is correct, and FALSE if it is incorrect.

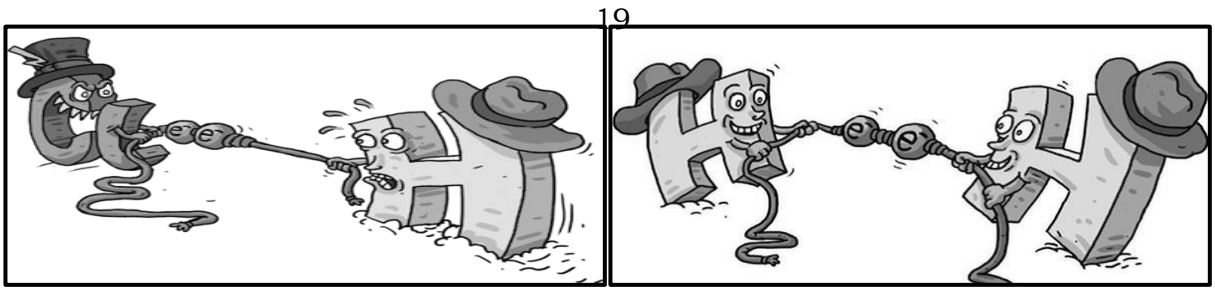
_____1. The polarity of a bond pertains to the distribution of electrical charge over the atoms joined by the bond.
_____2. Polarity arises from the difference in relative electronegativities of elements.
_____3. Polar covalent bond is a type of chemical bonding in which the electrons are shared equally between the atoms.
_____4. Non-polar covalent bonds unequally share electrons between the atoms.
_____5. Polarity of compounds does not affect its properties, like solubility.

Lesson 7 Polarity of Simple Molecules

What's New

Directions: Observe and analyze the picture below. What can we infer from these pictures?

19

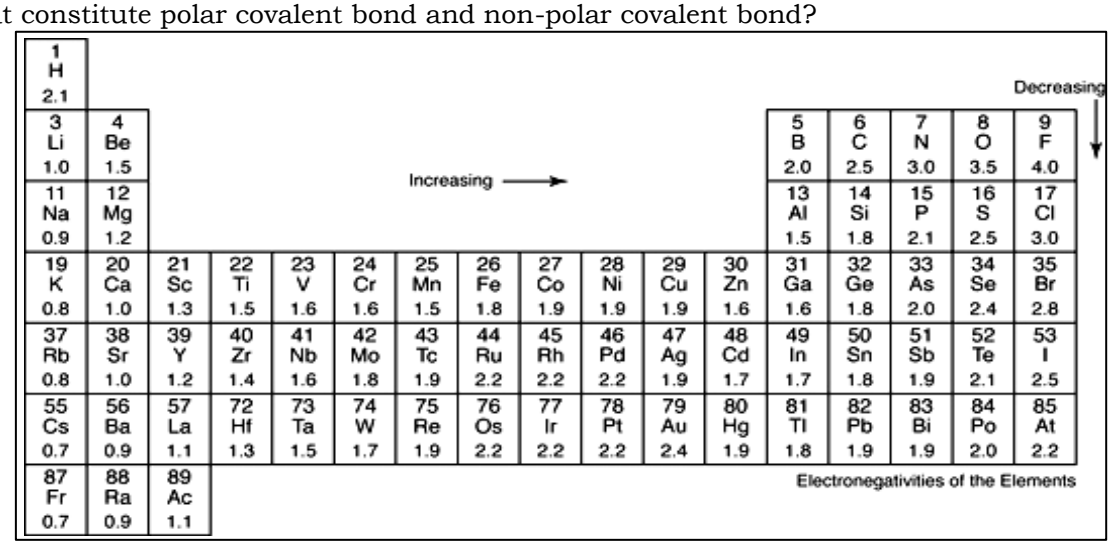


https://www.quora.com/What-is-the-simplest-way-to-distinguish-between-a-polar-and-non-polar-molecule

What Is It

The polarity of a bond pertains to the distribution of electrical charge over the atoms joined by the bond. It arises from the difference in relative electronegativities of elements. Electronegativity as discussed in the previous lessons is the power of an atom to attract electrons towards itself when it is in a compound. The polarity of a molecule can be classified as either a **polar molecule** that has polar bonds or **non-polar molecule** in which bonds are

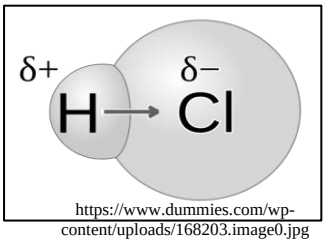
non-polar. **Polarity** affects various properties of a compound such as surface tension, solubility, melting point, and boiling point. Can you identify which of the two pictures represents a polar covalent bond? A non-polar covalent bond? What are the characteristics that constitute polar covalent bond and non-polar covalent bond?



1 H 2.1																
3 Li 1.0	4 Be 1.5											5 B 2.0	6 C 2.5	7 N 3.0	8 O 3.5	9 F 4.0
11 Na 0.9	12 Mg 1.2											13 Al 1.5	14 Si 1.8	15 P 2.1	16 S 2.5	17 Cl 3.0
19 K 0.8	20 Ca 1.0	21 Sc 1.3	22 Ti 1.5	23 V 1.6	24 Cr 1.6	25 Mn 1.5	26 Fe 1.8	27 Co 1.9	28 Ni 1.9	29 Cu 1.9	30 Zn 1.6	31 Ga 1.6	32 Ge 1.8	33 As 2.0	34 Se 2.4	35 Br 2.8
37 Rb 0.8	38 Sr 1.0	39 Y 1.2	40 Zr 1.4	41 Nb 1.6	42 Mo 1.8	43 Tc 1.9	44 Ru 2.2	45 Rh 2.2	46 Pd 2.2	47 Ag 1.9	48 Cd 1.7	49 In 1.7	50 Sn 1.8	51 Sb 1.9	52 Te 2.1	53 I 2.5
55 Cs 0.7	56 Ba 0.9	57 La 1.1	72 Hf 1.3	73 Ta 1.5	74 W 1.7	75 Re 1.9	76 Os 2.2	77 Ir 2.2	78 Pt 2.2	79 Au 2.4	80 Hg 1.9	81 Tl 1.8	82 Pb 1.9	83 Bi 1.9	84 Po 2.0	85 At 2.2
87 Fr 0.7	88 Ra 0.9	89 Ac 1.1														

We already learned that covalent compounds are characterized by covalent bonds that share electrons between two or more atoms. When these atoms differ in electronegativity or the ability of an atom to attract electrons toward itself, the bond polarity arises. Polarity pertains to the distribution of electrical charge over the atoms joined by the bond. In determining the polarity, one must know the electronegativity value of the elements present in a compound. The difference in electronegativity will tell you whether a compound has polar covalent bond or non-polar covalent bond. The number of atoms in a compound does not affect the difference in electronegativity. The electronegativity chart shows the electronegativity value of elements.

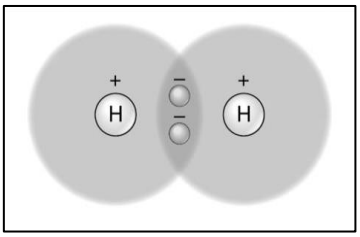
The **Polar covalent bond** is a type of chemical bonding in which the electrons are shared unequally between the atoms. In polar covalent bonds, one atom has a stronger electronegativity than the other atom thus attracts electrons. This causes the atom which has a stronger pull to carry a partial negative charge and the other atom carries a partial positive charge. If the difference in electronegativities is between 0.5 to 1.9, it is a polar covalent bond.



https://www.dummies.com/wp-content/uploads/168203.image0.jpg

Let's take HCl, hydrogen chloride as an example shown in Figure 2. The electronegativity of hydrogen is 2.1, chlorine on the other hand is 3.0. The difference is 0.9 thus HCl has a polar covalent bond. Since chlorine has higher electronegativity, it has a partial negative charge and electrons are attracted toward the chlorine atom, hydrogen on the other hand, has a partial positive charge.

Non-polar covalent bonds equally share electrons between the atoms. It occurs with covalent compounds that contain the same element or different elements which have very low difference in electronegativities. Non-polar covalent bond has 0.4 and below the difference in electronegativity. The figure shows a non-polar covalent bond between two hydrogen atoms because they equally share the electrons. Since they are the same atom, their electronegativity difference will equate to 0.



https://opentextbc.ca/anatomyandphysiology/wp-content/uploads/sites/142/2016/03/208_Covalent_Bond

Polarity of compounds greatly affects its properties, like solubility. The phrase "like dissolves like" means that only polar substances can dissolve another polar substances and non-polar substance can only dissolve non-polar substances as well.

What's More

Directions: Solve for the electronegativity difference of the following covalent compounds. Based on electronegativity difference, determine the polarity of the compound.

Compound	Difference in Electronegativity	Polarity
1. CO		

2. PCl_3		
3. HCl		
4. NH_3		
5. SO_2		

What I Can Do

Directions: Read and answer the question below.

Discuss how some compounds like mothballs cannot be dissolved in water even though it is considered as the "universal solvent". How is polarity involved in this situation?

_____

Assessment

Directions: Determine the polarity of the following compounds by solving the difference in electronegativity of the atoms.

1. H_2S _____
2. O_2 _____
3. C_6H_6 _____
4. HBr _____
5. $SeCl_2$ _____

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the topic about functional groups. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course, but the order in which you read them can be changed to correspond with the textbook you are now using.

At the end of this lesson, you are expected to:

Describe the different functional groups. (STEM_GC11OCIIg-j-87)

What I Know

Directions: Fill in the K-W-H-L Chart below about functional groups.

K	W	H	L
What do I know?	**What do I want to find out?**	**How can I found out what I learn?**	**What did I learn?**

Lesson 8 Functional Groups

What's New

Directions: Rearrange the letters of the correct term that is described by the corresponding statement.

_____ 1. ENESKLA – It refers to hydrocarbons containing one or more double bonds between the adjacent carbon atoms.

_____ 2. HERSTE – It consists of an oxygen atom and two carbon atoms forming single bonds.

_____ 3. COALSLOH – It is comprised of an oxygen atom bonded to one atom of hydrogen and one atom of carbon.

_____ 4. ITOLH – It comprises a sulfur atom that is bound to a hydrogen atom.
_____ 5. NESAMI – a nitrogen atom bound to a mixture of hydrogen and carbon.

What Is It

Alkenes and Alkynes

Alkenes are hydrocarbons containing one or more double bonds between the adjacent carbon atoms. Alkynes contain one or more triple bonds among adjacent carbon atoms. The presence of a double or triple bond result in non-alkane and alkyne reactivity (see figure below) undergoing characteristic reactions that will be dealt with later.

Figure 1: Alkenes (left) contain one or more double bonds while alkynes (right) contain one or more triple bonds.

Aromatics

An aromatic ring which is a six-carbon ring with alternative double bonds is the additional functional group that contains only carbon and hydrogen. The aromatic ring is also a ring representing the double bonds with a circle in the center. Several compounds, including steroids and drugs, contain aromatic rings (see figure below).

Figure 2: Aromatic rings contain alternating double bonds.

Alcohols

The functional alcohol group is comprised of an oxygen atom bonded to one atom of hydrogen and one atom of carbon. The carbon atom is meant to be part of a larger organic structure. The R−OH formula would be one way to indicate generic alcohol. R represents an organic fragment that directly bonds a carbon atom to the explicitly indicated functional group (in this case, OH). In general, the R group is a chain of carbon atoms.

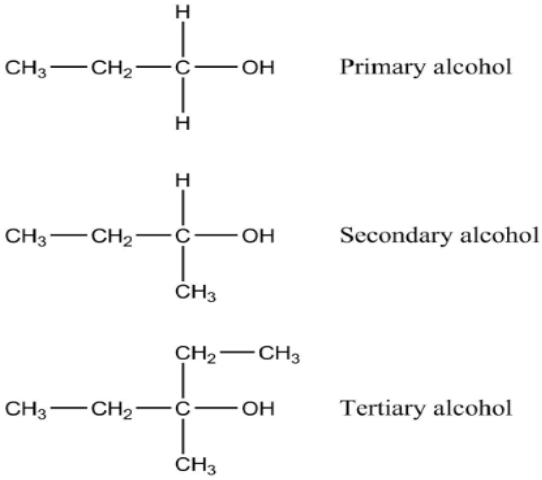


Figure 3: Primary, secondary, and tertiary alcohols.

Alcohols can be classified as primary, secondary, or tertiary, based on the carbon characteristics to which they are attached. The carbon directly bound to the oxygen atom in primary alcohol is also bound to precisely one carbon atom, with the other bonds usually going to the hydrogen atoms. In secondary alcohol, the carbon is attached to two other carbon atoms, and the carbon is bonded to three other carbon atoms in tertiary alcohol. The type of alcohol that is used will determine the product of some reactions. Note the naming of the alcohols as seen in the above chart. The −OH group location will be indicated with the carbon number to which it is attached.

Ethers

The ether functional group consists of an oxygen atom and two carbon atoms forming single bonds.

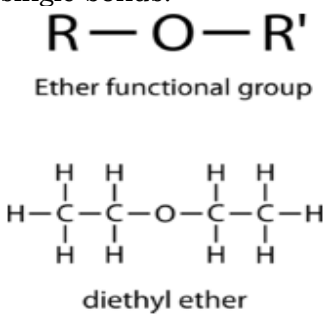


Figure 4: Ethers

Despite their low reactivity Ethers are suitable solvents for certain organic compounds. They dissolve the non-polar molecules readily. Diethyl ether is probably the ether that is most known. It is commonly used as a solvent and used as an inhalable anesthetic.

Thiol

The functional group of thiol comprises a sulfur atom that is bound to a hydrogen atom. It is very similar to a functional alcohol group with the sulfur substituting for O.

Figure 5: Thiol

Thiols are also called mercaptans, derived from the Latin phrase for "capturing mercury" due to the strong bonds it forms with compounds containing mercury. Some thiol compounds have a characteristic odor like rotten eggs. They are also applied to natural gas, which itself has no odor, as a way of detecting leaks because humans can smell its odor in very small quantities. The amino acid cysteine also comprises a thiol group which will be discussed later.

Amines

An amine is a nitrogen atom bound to a mixture of hydrogen and carbon.

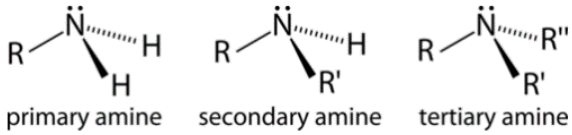


Figure 6: Amines

Amines can be classified as primary, secondary, or tertiary, just like alcohols. The rules for determining such types, however, are very different. The oxygen atom is usually bound to precisely one carbon atom in alcohol, and we are looking at the branching on the adjacent carbon, not at the oxygen atom itself. The nitrogen can be bonded to one, two, or three carbon atoms in a neutral amine, and this is how we decide whether it is called primary, secondary, or tertiary amine.

Aldehydes

The carbonyl, which is simply a carbon atom and an oxygen atom connected by a double bond, is a very common structural component of organic structures. Carbonyl reactivity is primarily dictated by the polarization of the C = O bond but the surrounding atoms also play a role in their specific pathways of reaction. While carbonyl is a component of several functional groups it is not a functional group in itself.

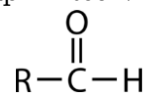


Figure 7: Aldehyde

An aldehyde is a carbonyl in which at least one hydrogen atom is bound to the carbon atom. The other carbonyl group may be an R-group or a hydrogen atom. Since the hydrogen atom is so small, it is very simple for other molecules to reach the partial positive charge on the carbonyl carbon making aldehydes a very reactive form of carbonyl. Aldehydes are versatile reactants for a wide variety of organic syntheses. Many aldehydes also have distinctive flavors and aromas. For example, the flavor of cinnamon is primarily due to the molecule cinnamaldehyde, and vanillin is the aldehyde most responsible for the smell and taste of the vanilla extract.

Ketones

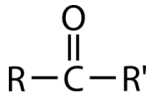


Figure 9: Ketone

A ketone requires a carbonyl in which the atom of carbon makes single bonds with two groups of Rs. Ketones are mostly subjected to the same reactions as aldehydes but tend to be slightly less reactive. The simplest ketone is acetone, where carbonyl carbon bond to two groups of CH_3.

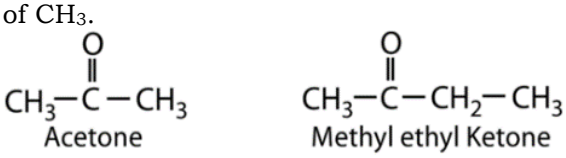


Figure 10: Two examples of ketones.

Carboxylic Acids

Carboxylic acids are another functional group of carbonyls, in which on one side the carbon atom is bound to an OH group and on the other, either phosphorus or a hydrogen atom.

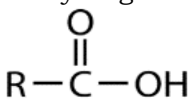


Figure 11: Carboxylic acid

The carboxylic acids are weak, as the name implies. When dissolved in water, and OH group that is directly linked to a carbonyl can ionize to a small degree. The explanation for this is that the resulting anion has relative stability. A carboxylate ion (see figure below), in which the negative charge is spread through resonance structures between two separate oxygen atoms, is more stable than an isolated oxygen-centric anion. The carboxylic acid is identical to the carboxylate ion. Carboxylate ions occur frequently in amino acids.

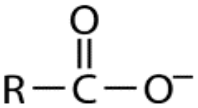


Figure 12: Carboxylate ion

Esters

An ester is similar to a carboxylic acid in that it contains a carbonyl where one additional oxygen atom and one carbon or hydrogen atom are bound to the carbon. The second oxygen atom, however, is bound to another carbon, rather than an acidic hydrogen atom. Carboxylic acids and esters are structurally related to each other in the same manner as alcohols and ethers.

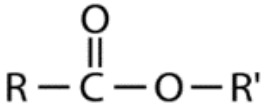


Figure 13: Ester

Esters can be formed in the presence of an acid catalyst by heating the carboxylic acids and alcohols. This cycle is reversible, and the starting materials can be regenerated in the presence of a weak base, by reacting an ester with water.

Amides

An amide is a carbonyl in which the nucleus is bound to one atom of nitrogen and one atom of nucleus or hydrogen. Alternatively, we may describe an amide as an amine in which a carbonyl is part of one of the carbon atoms attached to the nitrogen.

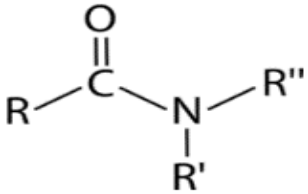


Figure 14: Amide

Haloalkanes

The haloalkanes, also known as alkyl halides, are a group of chemical compounds consisting of an alkane substituted by a halogen atom (Group 17 atom) with one or more hydrogens. There is a wide distinction between haloalkanes 'structural and physical properties and alkanes' structural and physical properties.

Figure 15: Haloalkane

What's More

Directions: Choose five functional groups and create a model of each using recyclable materials and your periodic table of elements.

What can you say about the models you created? List down all your observations and analysis.

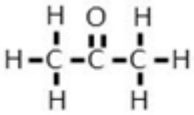
What I Can Do

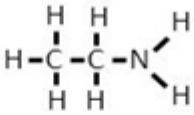
Directions: Read and answer the question below.

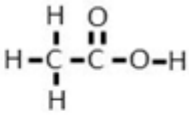
What do you think will happen if there are no functional groups?

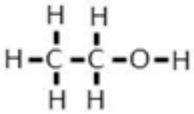
Assessment

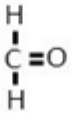
Directions: Encircle and name the functional groups found on the following organic compounds.

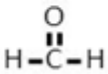
1. 

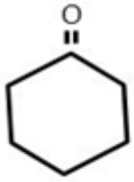
2. 

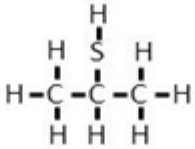
3. 

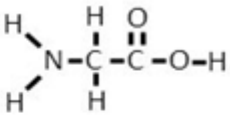
4. 

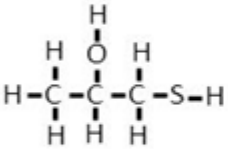
5. 

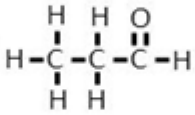
6. 

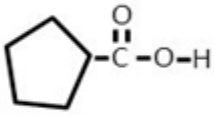
7. 

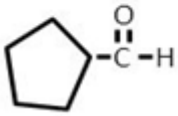
8. 

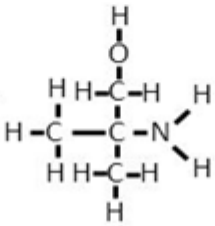
9. 

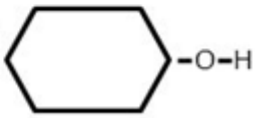
10. 

11. 

12. 

13. 

14. 

15. 

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the topic about structural isomerism. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course, but the order in which you read them can be changed to correspond with the textbook you are now using.

At the end of this lesson, you are expected to:

Describe structural isomerism, give examples. (STEM_GC11OCIIg-j-89)

What I Know

Directions: Match column A to column B.

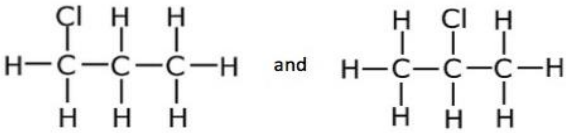
A	B
_____1. The isomers contain different functional groups in this variety of structural isomerism-that is, they belong to different compound families.	A. Chain Isomerism
_____2. The basic carbon skeleton remains unchanged in place isomerism but on that skeleton, essential groups are shifted around.	B. Position Isomerism
_____3. These isomers exist because of the branching possibilities in carbon chains.	C. Functional Group Isomerism

Lesson 9 Structural Isomerism

What's New

Directions: Read and answer the question below.

Analyze the given figure, what do you think is the relationship between the two organic molecules below?

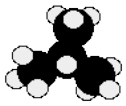


What Is It

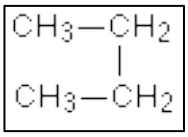
Chain Isomerism

These isomers exist because of the branching possibilities in carbon chains. There are two butane isomers, for example, C_4H_{10}. The carbon atoms in one of them lie in a "straight line" and in the other the line is branched.

$CH_3—CH_2—CH_2—CH_3$



$CH_3—CH(CH_3)—CH_3$

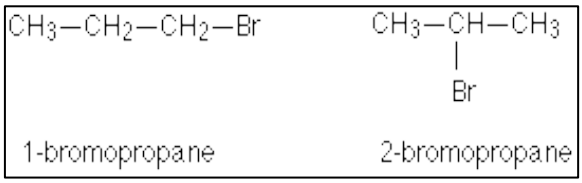


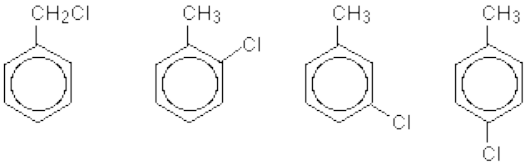
Be careful not to draw "false" isomers that are the original molecule's distorted versions. This structure, for example, is just the straight chain form of rotated butane around the central carbon-carbon bond.

Position isomerism

The basic carbon skeleton remains unchanged in place isomerism but on that skeleton, essential groups are shifted around.

For example, the molecular formula C_3H_7Br has two structural isomers. In one of them, the bromine atom is at the end of the chain and in the other the bromine atom is connected in the center.



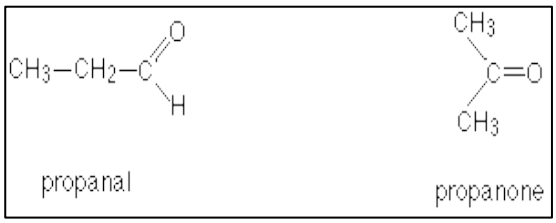


For benzene rings, you can also get isomers for a place. Find the C_7H_7Cl molecular formula. There are four different isomers that you can create based on the chlorine atom's position. For one case it is bound to the carbon atom of the side-group, and then there are three other potential locations around the ring-next to the CH_3 group, next-but-one to the CH_3 group, or opposite to the CH_3 group.

Functional group isomerism

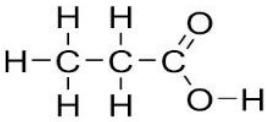
The isomers contain different functional groups in this variety of structural isomerism-that is, they belong to different compound families (various homologous series).

A molecular C_3H_6O formula may be either propanal (an aldehyde) or propanone (a ketone).



What's More

Directions: Draw all the possible structural isomers of the organic compound below.



What I Can Do

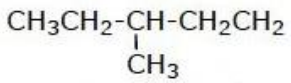
Directions: Read and answer the question below.

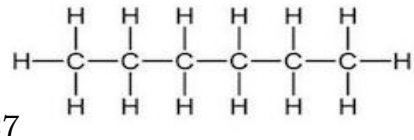
What is the importance of structural isomers in organic compounds?

Assessment

Directions: Read and analyze the question below.

All of the following are chain isomers of each other, except one. Which is the exception and explain why?





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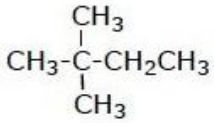
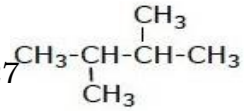




27

Lesson 10 Simple Reactions of Organic Compounds

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the topic about simple reaction of organic compounds. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course, but the order in which you read them can be changed to correspond with the textbook you are now using.

At the end of this lesson, you are expected to:

Describe some simple reactions of organic compounds, combustion of organic fuels, addition, condensation, and saponification of fats. (STEM_GC11OCIIg-j-90)

What I Know

Directions: Fill in the K-W-H-L Chart below about simple reactions of organic compounds.

K	W	H	L
What do I know?	**What do I want to find out?**	**How can I found out what I learn?**	**What did I learn?**

What's New

Directions: Rearrange the letters of the correct term that is described by the corresponding statement.

_____ 1. SUBMOBNOTI – It a reaction in which a material reacts with oxygen gas in the form of light and heat, releasing energy.

_____ 2. TIONDIDA – a reaction in which an unsaturated molecule is added to an atom or molecule, producing more product.
_____ 3. DENCONSATION – It a reaction that combines two molecules to form a single molecule.
_____ 4. POSANICAFINOTI – describes the reaction of an ester with alkaline hydrolysis.

What Is It

Combustion

A combustion reaction is a reaction in which a material reacts with oxygen gas in the form of light and heat, releasing energy. Combustion reactions must have O_2 as one reactant. Hydrogen gas combustion produces water vapor.

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(g)$$

Addition

The addition reaction is a reaction in which an unsaturated molecule is added to an atom or molecule, producing more product. An additional reaction can also be described as inserting a molecule across an alkene's double bond, or across an alkyne's triple bond. Knowing that "*ation*" means to add, it would make sense to assign common names to such reactions, such as hydrogenation, hydration, or chlorination. Remember that the processes of hydrogenation (H_2 adding) and hydration (H_2O adding) are somewhat different.

One additional form of reaction is called hydrogenation. Hydrogenation is a reaction that happens when molecular hydrogen is added to an alkene to create an alkane, or when an alkane or alkane is added to an alkyne. Usually, the reaction is conducted by using a transition metal catalyst. For example, ethene reacts to the ethane forming with hydrogen.

$$CH_2 = CH_2(g) + H_2(g) \rightarrow CH_3CH_3(g)$$

Note that the hydrogenation reaction is also a redox reaction. Ethene is reduced because the oxidation numbers of the carbon atoms change from −2 to −3 as a result of the reaction.

Condensation

A condensation reaction is a reaction that combines two molecules to form a single molecule. A small molecule is normally removed during a condensation reaction, often with water. Amino acids are essential biological molecules with a functional amine group on one end of the molecule, and a functional carboxylic acid group on the other. As two amino acids interact in a condensation reaction, the amine nitrogen of one amino acid and the carboxylic carbon of the second amino acid form a covalent bond. A water molecule is then taken out as a second element (see figure below).

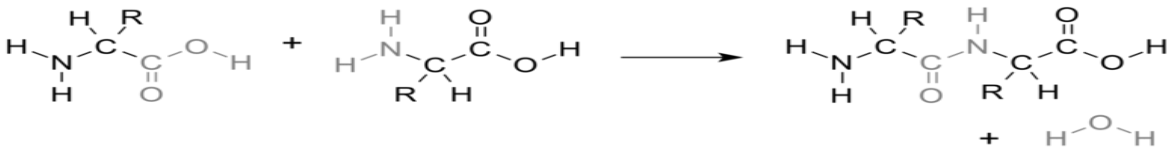


Figure 1: Amino acids join together to form a molecule called a dipeptide. The −OH from the carboxyl group of one amino acid combines with a hydrogen atom from the amine group of the other amino group to produce water (blue).

Saponification

Saponification describes the reaction of an ester with alkaline hydrolysis. The term saponification originally described the hydrolysis of long-chain esters to produce soap molecules called fatty acid esters, which are the salts of fatty acids. One such molecule of soap is sodium stearate that is formed from ethyl stearate hydrolysis.

$$C_{17}H_{35}COOC_2H_5 + NaOH \rightarrow C_{17}H_{35}COO^{-}Na^{+} + C_2H_5OH$$

ethyl stearate sodium stearate (soap)

What's More

Directions: Complete the table below by giving different examples of organic reactions that undergo combustion, addition, condensation, and saponification.

Combustion	Addition

Condensation	Saponification

Follow-up Questions:

1. How will you differentiate the four reactions of organic compounds?

_____

2. Which among the four reactions can be easily destroyed by another reaction?

_____

What I Can Do

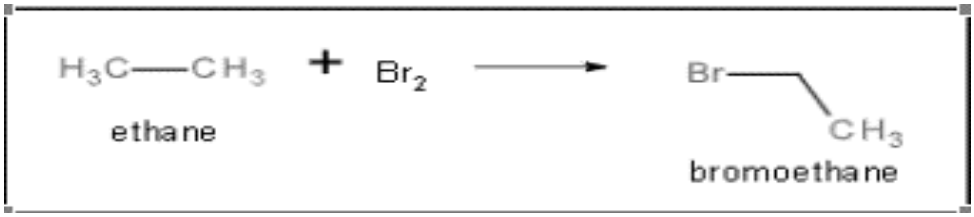
Directions: Read and answer the question below.

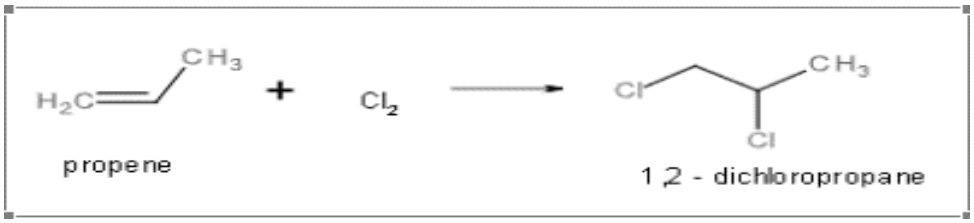
What do you think is the importance of organic reactions in our everyday life?

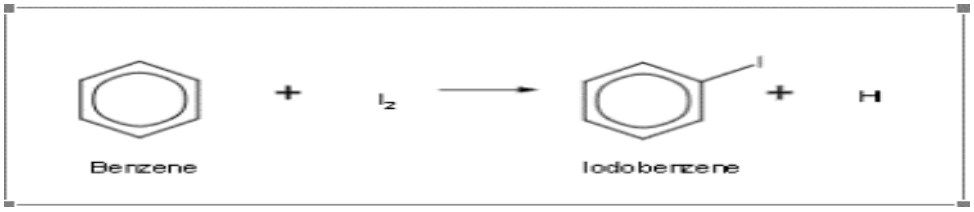
_____

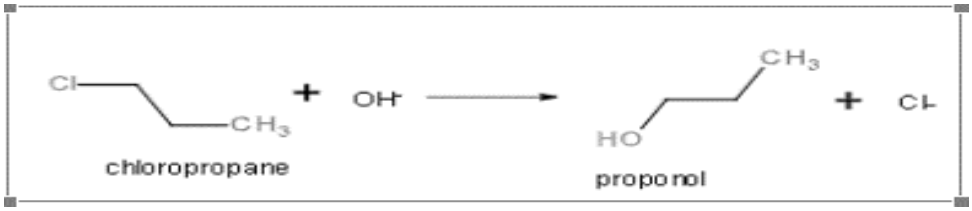
Assessment

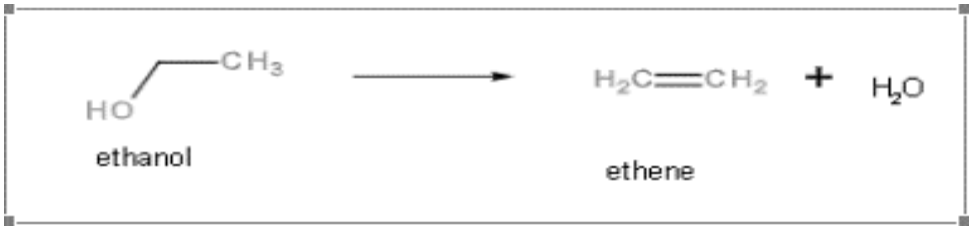
Directions: Choose from the following the type of reaction shown in each number. (Addition, elimination, substitution, oxidation, reduction, and condensation).

1. 

2. 

3. 

4. 

5. 

Lesson 11 Formation of Polymers

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the topic about formation of polymers. The scope of this module permits it to be

used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course, but the order in which you read them can be changed to correspond with the textbook you are now using.

At the end of this lesson, you are expected to:

1. **Describe the formation of polymers. (STEM_GC11OCIIg-j-91)**
2. **Explain the properties of some polymers in terms of their structure. (STEM_GC11OCIIg-j-93)**

What I Know

Directions: Match the correct prefix or suffix or definition to its meaning/word. Choose from the box provided below.

dehydrate	hydrolysis	synthesis	monomer	polymer

1. To make something _____.
2. Many monomers hooked together make a _____.
3. Means to lose or remove water; to take water away _____.
4. Means water (as in gaining water) _____.
5. Building block or single unit of a polymer is a _____.

What's New

Directions: Arrange the different steps of addition polymerization.

_____ Propagation. The radical will react with another ethylene molecule monomer.
_____ Termination. The process gets terminated when two long-chained radicals combine to form the polyethylene polymer.
_____ Initiation. An initiator molecule is heated to produce free radicals.

What Is It

How are Polymers Made?

The chemical reaction by which the monomers are linked together to form polymers is called polymerization. There are several types of polymerization reactions. The basic types are the addition polymerization and the condensation polymerization reactions.

1. Addition Polymerization

In addition to polymerization, the entire monomer becomes part of the polymer. They involve molecules with double bonds or triple bonds. Consider the formation of polyethylene, a stable polymer used widely as packaging wrap. The polymerization reaction consists of three steps:

$$R_2 \longrightarrow 2\,R\bullet$$

Step 1: Initiation An initiator molecule is heated to produce free radicals. These free radicals have one unpaired electron each and are therefore very reactive

$$R\bullet + CH_2 = CH_2 \longrightarrow R-CH_2-CH_2\bullet$$

seeking other electrons from other molecules to pair up with. The free radicals react with the ethylene molecule monomer breaking the double bond and forming a new radical.

Step 2: Propagation The radical will react with another ethylene molecule monomer. This is repeated many times over and a long chain is formed.

$$R-CH_2-CH_2\bullet + CH_2=CH_2 \longrightarrow R-CH_2-CH_2-CH_2-CH_2\bullet$$

Step 3: Termination The process gets terminated when two long-chained radicals combine to form the polyethylene polymer.

$$R-(CH_2-CH_2)_n-CH_2-CH_2\bullet + R-(CH_2-CH_2)_n-CH_2-CH_2\bullet \longrightarrow$$
$$R-(CH_2-CH_2)_n-CH_2-CH_2-CH_2-CH_2-(CH_2-CH_2)_n-R$$

where $(CH_2 — CH_2)n$ is a shorthand convention that represents n repeating units.

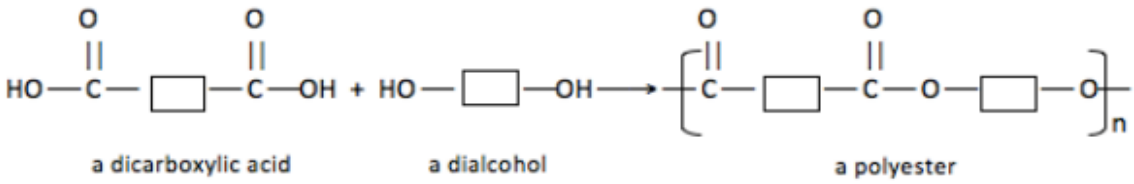
Polyethylene is an example of a homopolymer. This is a type of polymer where there is only one type of monomer. Other examples of monomers used to produce polymers are shown in the table.

In the examples given above, ethylene ($CH_2 = CH_2$) and tetrafluoroethylene ($CF_2 = CF_2$) are **symmetric monomers** (the carbons have the same substituents) while vinyl chloride, styrene, and propene are **asymmetric monomers** (the carbons in the monomer have different substituents). The examples (polyethylene, polystyrene, polypropylene, and Teflon) are synthetic polymers.

2. Condensation Polymerization

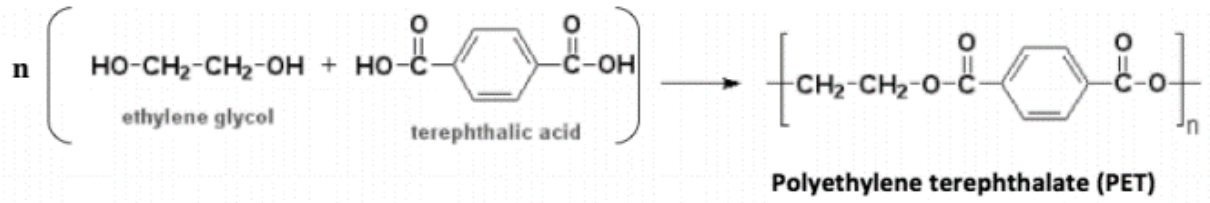
Condensation polymers are those formed through a condensation reaction; that is, where monomers join together at the same time losing a small molecule like water as a by-product.

Recall that when a carboxylic acid reacts with alcohol, an ester is formed. Now when a monomer that has two carboxylic acid groups, one at each end, is made to react with a second monomer containing two –OH groups, one at each end, then many ester linkages are formed resulting in a polyester.



The reaction of a dicarboxylic acid and a dialcohol to produce polyester. If the dicarboxylic acid is made to react with a diammine (instead of a dialcohol), then a **polyamide** is formed such as nylon.

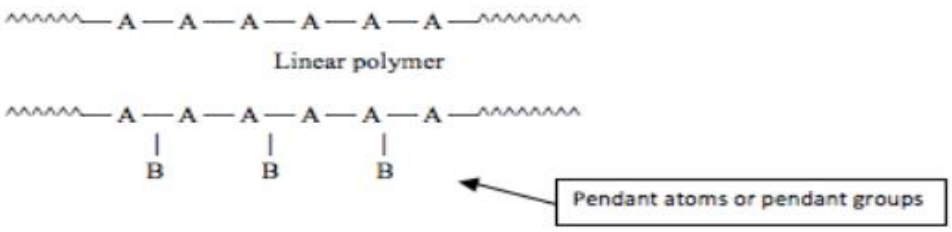
For example, the polymer polyethylene terephthalate (PET or sometimes called PETE) is formed by the reaction of terephthalic acid and ethylene glycol. PET is a polyester.



Polymer Arrangements and Structures

Polymers can be arranged several ways. The arrangement of the polymer chains affects their properties such as whether they are stiff or rigid, crystalline or amorphous.

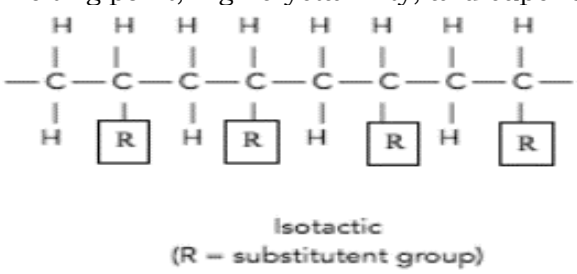
A **linear polymer** is one where the arrangement of atoms is like that of a long chain. This long-chain is often referred to as the backbone. Atoms or small groups of atoms attached to the long chain are called pendant atoms.

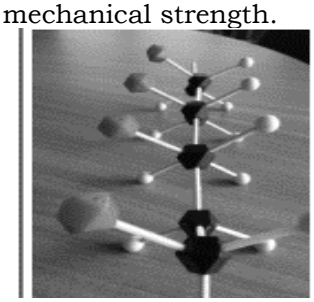


The long chains are not stiff; they are flexible. They twist and bend and wrap around one another. They become entangled like strands of spaghetti, especially in the solid state. This makes the polymer strong. When polymers are dissolved in solvents, the chains move very slowly compared to small molecules. This is due to the entanglement of the long-chain strands. Thus, polymers dissolved in solvents can be more viscous than pure solvents.

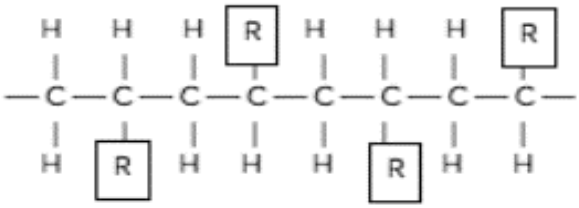
The arrangement of the pendant atoms or pendant groups attached to the backbone gives different properties to the polymer. Three distinct arrangements are observed:

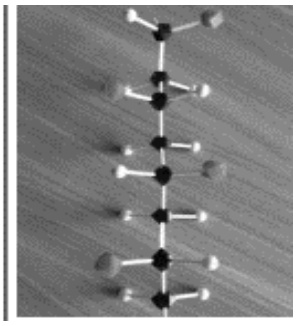
a. The **isotactic** arrangement is where all the pendant groups or substituents (represented by R —) are on the same side of the polymer chain. They pack efficiently resulting in polymers with a high melting point, high crystallinity, and superior mechanical strength.



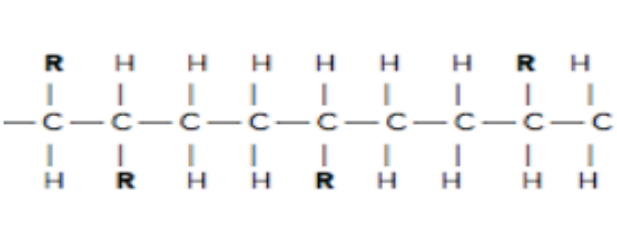


b. A **syndiotactic polymer chain** is one where the substituent group alternates from left to right of the asymmetric carbons. They pack less efficiently than isotactic chains.





c. In **atactic polymers**, the substituents occur randomly. Therefore, they do not pack well. These polymers are rubbery, not crystalline, and relatively weak.



What's More

Directions: Collect at least 20 types of polymeric materials found at your home. Then, complete the table below.

Polymeric Material Code	Frequency (count)	Percentage	Describe Uses or Applications Collected
1			
2			
3			
4			
5			
6			
7			
Total			

Hint: *Plastics are identified by numbers 1 through 7. Usually located on the bottom of the package, this number is code for the specific type of plastic you're holding. Here are a couple of examples of the identification numbers and their corresponding products:*

- *#1 PET (Polyethylene terephthalate) – Fizzy drink bottles, oven-ready meal trays and water bottles*
- *#2 HDPE (High-density polyethylene) – Milk bottles, detergent bottles, yogurt and margarine tubs, cereal box liners and grocery, trash, and retail bags*
- *#3 PVC (Polyvinyl chloride) – Cling film (plastic food wrap), vegetable oil bottles, loose-leaf binders and construction products such as plastic pipes*
- *#4 LDPE (Low-density polyethylene) – Dry cleaning bags, produce bags, trash can liners, bread bags, frozen food bags, and squeezable bottles, such as mustard and honey*
- *#5 PP (Polypropylene) – Ketchup bottles, medicine bottles, aerosol caps, and drinking straws*
- *#6 PS (Polystyrene) – Compact disc jackets, grocery store meat trays, egg cartons, aspirin bottles, foam packaging peanuts, and plastic tableware*
- *#7 Other - Three- and five-gallon reusable water bottles, certain kinds of food* containers and Tupperware

Guide Questions:

1. Which polymeric material was most commonly collected at your home?

_____

2. Which polymeric material was the least commonly collected?

_____

3. What percentage of polystyrene materials were collected?

_____

4. What was the most common application collected for polyethylene terephthalate materials collected?

_____

5. Ranking the polymeric materials collected from the most common to least common, where did high-density polyethylene rank in your table?

_____

What I Can Do

Directions: Read and answer the question below.

What is the importance of polymers in our everyday life?

_____

Assessment

Directions: Encircle the letter of the best answer.

1. A large molecule that consists of many repeating units is called a
   A. Polymer
   B. Monomer
   C. Polypeptide
   D. Polystyrene
   E. Polyethylene
2. Each unit that comes together to form a polymer is called a ___.
   A. ethylene
   B. monomer
   C. copolymer
   D. thermoset
   E. amino acid
3. The reactions in which polymers are produced are called _____ reactions.
   A. synthesis
   B. oxidation
   C. decomposition
   D. polymerization
   E. single displacement
4. Which of the following is a requirement for monomers to be able to undergo addition polymerization?
   A. The presence of a lone pair
   B. The presence of a nitrogen atom
   C. The presence of an oxygen atom
   D. The presence of an electronegative atom
   E. The presence of a carbon-to-carbon double bond
5. In a _____ reaction for the synthesis of a polymer, a small molecule is removed from the reacting materials and produced as a by-product.
   A. addition
   B. oxidation
   C. elimination
   D. condensation
   E. substitution

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the topic about biomolecules and its functions. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course, but the order in which you read them can be changed to correspond with the textbook you are now using.

At the end of this lesson, you are expected to:

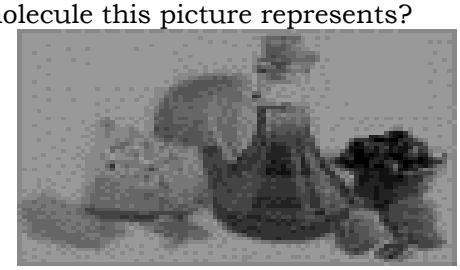
Describe the structure of proteins, nucleic acids, lipids, and carbohydrates, and relate them to their function. (STEM_GC11OCIIg-j-95)

What I Know

Directions: Encircle the letter of the best answer.

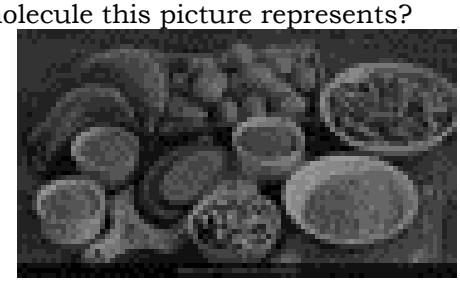
1. One function of carbohydrates is to_____.
   A. maintains blood flow
   B. catalyzes physiological reactions
   C. store and transport genetic material
   D. provides structural framework of cell walls
2. The three classifications of carbohydrates are _____.
   A. sucrose, maltose, and lactose
   B. starch, glycogen, and cellulose
   C. triglycerides, steroids, and waxes
   D. monosaccharide, disaccharide, and polysaccharide
3. Which elements make up a carbohydrate?
   A. hydrogen, oxygen, and carbon
   B. calcium, potassium, and oxygen
   C. hydrogen, nitrogen, and oxygen
   D. carbon, phosphorus, and oxygen

4. What type of biomolecule this picture represents?



A. lipid B. protein C. nucleic acid D. carbohydrates

5. What type of biomolecule this picture represents?



A. lipid B. protein C. nucleic acid D. carbohydrates

Lesson 12 Biomolecules and Its Function

What's New

Directions: Read and answer the question below.

What is the importance of understanding the different biomolecules in our life?

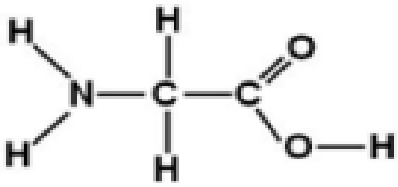
_____

What Is It

I. Proteins

Proteins are natural polymers. They are very large molecules that are critical for the functions of the human body. They are made from the linkage of monomers called **amino acids**. Notice that amino acids have two functional groups. a. Carboxylic group (—COOH) and b. Amine group (—NH2).

There are 20 kinds of amino acids depending on the —R group. The simplest amino acid is glycine where R is a hydrogen atom. The body cannot make all the amino acids required by the body and is dependent on protein taken through food.



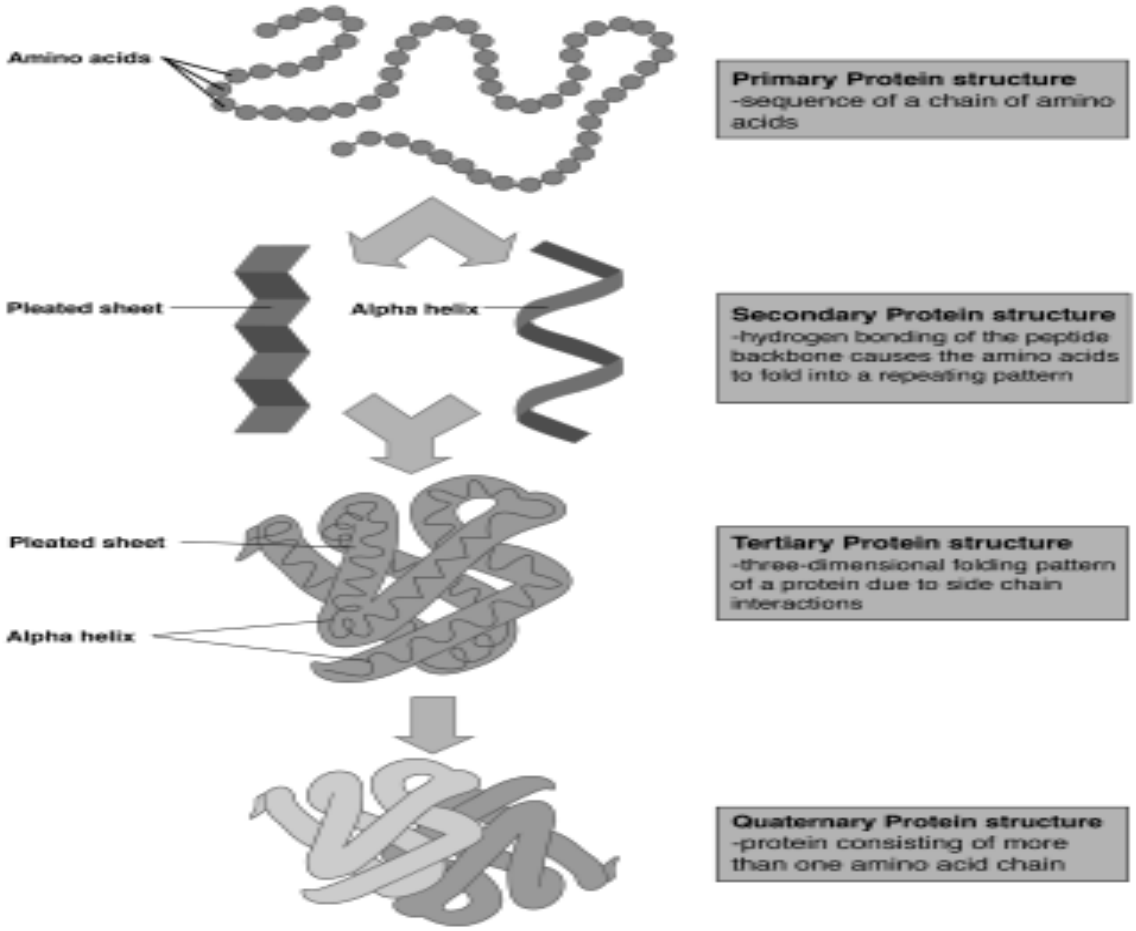
Different Levels of Protein Structure

The sequence and the structure of proteins are important in determining their functions. There are four levels of protein structures: the **primary** structure, **secondary** structure, **tertiary** structure, and **quaternary** structure.

The **primary** structure refers to the linear sequence of amino acids joined by peptide bonds such as the sequence of amino acids below.

The single bonds in the polypeptide chain allow rotation. Therefore, the polypeptide chain can twist and fold in a variety of ways. These folded structures are referred to as **secondary** protein structures. The two essential secondary structures are the **alpha helix** and the **beta-pleated sheets**. The structures are stabilized by hydrogen bonds between

amino acids. In the alpha-helix structure, the chain twists like a corkscrew while the chain takes the form of a folded sheet in the beta-pleated structure. Several secondary structures come together forming tertiary structures. When several **tertiary structures** come together, a **quaternary protein structure** is formed. For example, the protein hemoglobin is a quaternary structure formed by four tertiary structures.



Protein Functions

Proteins are important molecules in a cell. Each protein in the body has a specific function. Some of the types of proteins and their functions are:

a. Antibodies - proteins involved in defending the body against antigens. They are the molecules of the immune system.
b. Contractile proteins - responsible for body movement such as muscle contraction
c. Enzymes - proteins that catalyze (speed up) or facilitate biochemical reactions
d. Hormonal proteins - serve as messenger proteins to help coordinate some body functions. An example is insulin (which controls blood sugar concentration).
e. Structural proteins - are fibrous and provide support. An example is collagen which provides support to connective tissues.
f. Storage proteins - store amino acids like casein in milk.
g. Transport proteins - are carrier proteins that move molecules from one place to another in the body. An example is a hemoglobin which transports oxygen.

II. Nucleic Acids

Nucleic acids are natural polymers with very large molar masses. The two main types of nucleic acids are **deoxyribonucleic acid (DNA)** and **ribonucleic acid (RNA).** DNA and RNA are polymers made up of monomers in the form of nucleotides. When these nucleotides combine, they form polynucleotides. DNA carries the genetic information passed on from parents to children. RNA plays an important role in protein synthesis.

Features of DNA and RNA

	DNA	RNA
Function	Repository of genetic information	Involved in protein synthesis
Structure	Double helix	Usually single-strands
Sugar used	deoxyribose	Ribose
Bases used	C,T,A,G	C, U, A, G

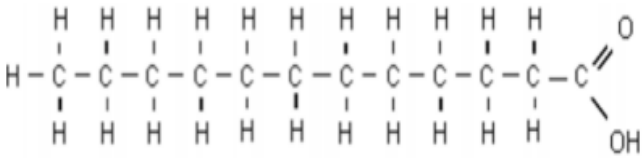
III. Lipids

Lipids are a class of naturally occurring organic compounds distinguished by their solubility in an organic solvent (and not in water). Lipids are hydrophobic, nonpolar, and made up mostly of hydrocarbon chains. Some of the more important lipids are triglycerides (fats), saturated and unsaturated fatty acids, waxes, phospholipids, and steroids. Some of these are further discussed below.

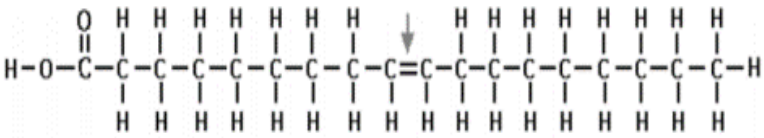
1. Fatty Acids

Fatty acids consist of a long hydrocarbon chain (typically about 12 – 18 carbons) attached to a carboxyl group.

a. **Saturated fatty acid** – contains only single C-C bonds because the carbon atoms are saturated or filled up with hydrogens. Because their structure is straight, they can pack well and are solid at room temperature (e.g., fat in butter).



b. **Unsaturated fatty acids** – contain carbon-carbon double bonds. When there is only one C-C double bond, it is called **monounsaturated**; if there are several C-C double bonds, they are called **polyunsaturated**. Remember that when there are double bonds, there will be geometric isomers (cis and trans). Because of the double bonds, they do not pack as tightly as saturated fatty acids. They are usually liquids at room temperature. An example of an unsaturated fatty acid is olive oil.



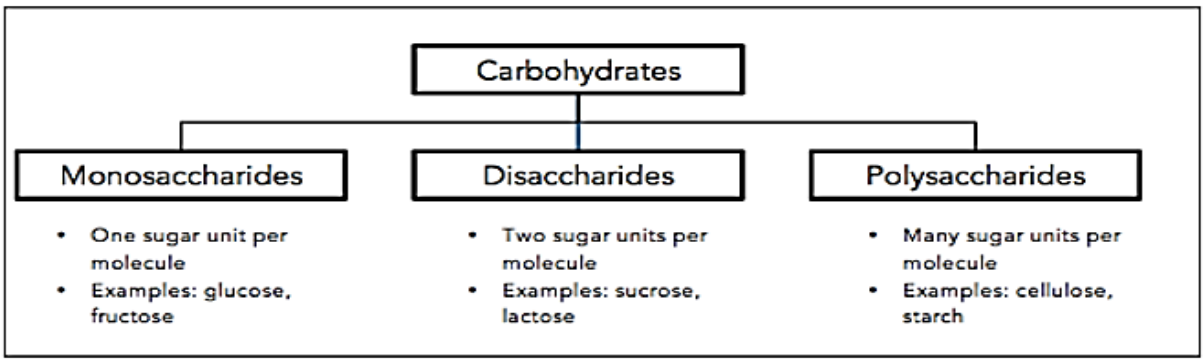
Oleic Acid- Monounsaturated Fatty Acid

c. **Fats and Oils** - Fat molecules have two parts: a glycerol backbone and three fatty acid chains. The resulting molecule is called **a triglyceride**.

IV. Carbohydrates

Carbohydrates are compounds made up of carbon, hydrogen, and oxygen. They are also known as saccharides. They have the general formula Cx (H_2O). Carbohydrates function as the energy source of the body.

A simple general classification of carbohydrates is according to the number of sugar units (saccharides) present in the molecule: **monosaccharides**, **disaccharides**, and **polysaccharides.**



Classification of carbohydrates

What's More

Directions: Encircle the letter of the best answer.

1. What elements make up a carbohydrate?
   A. hydrogen, calcium, oxygen C. hydrogen, carbon, oxygen
   B. carbon, potassium, oxygen D. carbon, magnesium, hydrogen
2. _____ is known as "animal starch".
   A. Glucose B. Cellulose C. Fructose D. Glycogen
3. _____ is the monosaccharide found in starch.
   A. Glucose B. Fructose C. Maltose D. Lactose
4. What biomolecule do enzymes belong to?
   A. Carbohydrate B. Lipid C. Nucleic Acid D. Protein
5. Which biomolecule is found in pasta and bread?
   A. Lipid B. Protein C. Nucleic Acid D. Carbohydrate

What I Can Do

Directions: A. Complete the paragraph below.

Nucleic acids are the organic compounds that control the making of _____ and contain the _____, _____ to pass to the next generation.

B. Read and answer the question below.

1. Why are there so many different shapes of proteins?

_____

2. What atoms make up carbohydrates?

_____

3. What are the two types of nucleic acids in the body? Explain each.

_____

4. What causes fatty acids to be saturated and unsaturated?

_____

Assessment

Directions: Encircle the letter of the best answer.

1. Which biomolecule is made of nucleotides and contains your DNA?
   A. Lipids B. Protein C. Carbohydrate D. Nucleic Acid
2. Which biomolecule helps form muscles?
   A. Lipids B. Protein C. Carbohydrate D. Nucleic Acid
3. Which biomolecule is responsible for insulation and energy storage?
   A. Lipids B. Protein C. Carbohydrate D. Nucleic Acid
4. Which biomolecule has sugars and starches?
   A. Lipids B. Protein C. Carbohydrate D. Nucleic Acid
5. Which biomolecule is found in fats, oils, and waxes and is a source of long-term energy?
   A. Lipids B. Protein C. Carbohydrate D. Nucleic Acid

REFERENCES

Lesson 1 and 2
https://courses.lumenlearning.com/introchem/chapter/quantum-numbers/
https://chemed.chem.purdue.edu/genchem/topicreview/bp/ch6/quantum.html
https://www.angelo.edu/faculty/kboudrea/general/quantum_numbers/Quantum_Numbers.htm
https://byjus.com/chemistry/quantum-numbers/
https://courses.lumenlearning.com/cheminter/chapter/electron-configurations/
https://www.chem.fsu.edu/chemlab/chm1045/e_config.html
https://courses.lumenlearning.com/introchem/chapter/diamagnetism-and-paramagnetism/
Chang, Raymond. Chemistry 6th Edition.McGraw-Hill. Unites States of America

Lesson 3 and 4
https://byjus.com/chemistry/electron-configuration/
https://www.khanacademy.org/science/class-11-chemistry-india/xfbb6cb8fc2bd00c8:in-in-structure-of-atom/xfbb6cb8fc2bd00c8:in-in-electronic-configuration-of-atoms/a/electron-configurations-article
https://chem.libretexts.org/Bookshelves/Physical_and_Theoretical_Chemistry_Textbook_Maps/Supplemental_Modules_(Physical_and_Theoretical_Chemistry)/Electronic_Structure_of_Atoms_and_Molecules/Electronic_Configurations/Electronic_Configurations_Intro
https://www.sgasd.org/cms/lib/PA01001732/Centricity/Domain/121/Practice%20orbtial%20notation%20answers.pdf
http://content.njctl.org/courses/science/chemistry/atomic-structure/atomic-structure-multiple-choice-review-2/atomic-structure-and-electron-configurations-multiple-choice-review-2015-09-061.pdf
Chang, Raymond (1998). Chemistry 6th Edition. McGraw-Hill Companies, Inc. United States
Padolina, Ma. Christina, et al (2004). Conceptual and Functional Chemistry: Modular Approach. Vibal Publishing House, Inc. Quezon City, Philippine
Chang, R. (2014). General Chemistry: Essential Concepts (7th ed.). New York: McGraw Hill Co. Inc.
Wilbraham, A. G., Staley, D. D., Matta, M. S. and Waterman, E. L. (2003). Chemistry (5th ed.). Singapore: Pearson Education (Asia) Pte. Ltd.

Lessons 5 to 7
Boundless Chemistry. (2020, June 29). Retrieved from Lumen Learning: https://courses.lumenlearning.com/boundless-chemistry/chapter/molecular-geometry/
Daquigan, M. (2016). General Chemistry 1.
Geometry of Molecules. (2020, August 20). Retrieved from Chemistry Libretexts: https://chem.libretexts.org/Bookshelves/Physical_and_Theoretical_Chemistry_Textbook_Maps/Supplemental_Modules_(Physical_and_Theoretical_Chemistry)/Chemical_Bonding/Lewis_Theory_of_Bonding/Geometry_of_Molecules
Morsch, L. (2020, June 29). Polarity of Molecules. Retrieved from Chemistry Libretexts: https://chem.libretexts.org/Courses/University_of_Illinois_Springfield/UIS%3A_CHE_267_-_Organic_Chemistry_I_(Morsch)/Chapters/Chapter_01%3A_Structure_and_Bonding/1.12%3A_Polarity_of_Molecules
The Polarity of Molecules. (2020, June 29). Retrieved from Britannica: https://www.britannica.com/science/chemical-bonding/Resonant-structures

Lessons 8 to 10
https://www.masterorganicchemistry.com/2010/10/06/functional-groups-organic-chemistry/#:~:text=Functional%20groups%20are%20specific%20groupings,acids%2C%20ketones%2C%20and%20ethers.
https://chem.libretexts.org/Courses/University_of_Kentucky/UK%3A_CHE_103_-_Chemistry_for_Allied_Health_(Soult)/Chapters/Chapter_4%3A_Structure_and_Function/4.4%3A_Functional_Groups
https://chem.libretexts.org/Courses/University_of_Kentucky/UK%3A_CHE_103_-_Chemistry_for_Allied_Health_(Soult)/Chapters/Chapter_12%3A_Organic_Reactions/12.1%3A_Organic_Reactions
https://www.google.com/search?q=worksheets+on+functional+group&authuser=1&source=lnms&tbm=isch&sa=X&ved=2ahUKEwiYzdHw_dvqAhVRyYsBHd-UDtEQ_AUoAXoECA0QAw&biw=676&bih=676#imgrc=W0G4VKzeX0cRjM

https://www.google.com/search?q=worksheets+simple+reactions&tbm=isch&ved=2ahUKEwjdpqfDj9zqAhXTwIsBHSnnBnsQ2-cCegQIABAA&oq=worksheets+simple+reactions&gs_lcp=CgNpbWcQAzoCCAA6BAgAEB46BggAEAUQHjoGCAAQCBAeULSWaViZtmlgirhpaABwAHgAgAFNiAGaCJIBAjE2mAEAoAEBqgELZ3dzLXdpei1pbWfAAQE&sclient=img&ei=9LQVX93cJtOBr7wPqc6b2Ac&authuser=1&bih=686&biw=1381&hl=en&hl=en#imgrc=woe8Go0RX3ut9M

Lessons 11 to 12
https://studylib.net/doc/6598406/worksheet--carbohydrates---manhasset-public-schools
https://www.teachengineering.org/lessons/view/csu_polymer_lesson01

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